

Generating HOMO/LUMO Orbital GIFs

Once you have gathered the perpendicular and parallel rotations for your molecule, you can start.

1. Open the Required Directories

Begin by opening:

- The GitHub repository containing the video creation files.
- Your folder containing the parallel (`para`) or perpendicular (`perp`) rotations.

Without loss of generality, the following instructions will refer only to the `para` files, but the same procedure applies to the `perp` files.

2. Generate Molden Files

You must create `.molden` files that will be used by Jmol to generate your orbitals using the `.gbw` files.

Copy the file named `conv_molden` from the Git repository into the `para` rotations directory.

3. Convert ORCA Outputs to Molden Format

In the terminal, navigate to the directory containing the `para` files and run:

```
bash conv_molden
```

This will create Molden files for each of your log files.

Next, create a directory to store the Molden files:

```
mkdir MoldenFiles
```

Move the generated files into it:

```
mv *.molten.output MoldenFiles
```

Now navigate into the `MoldenFiles` directory and add the following files from the Git repository:

```
make_frames  
conv_png  
make_gif.py
```

Create two directories by running:

```
mkdir homo  
mkdir lumo
```

Finally, place your copy of **Jmol** into the `MoldenFiles` directory.

Note: Jmol version **16.3.27** is a known working version; we've had issues with newer versions of Jmol.

4. Select HOMO or LUMO Generation

Before running any programs, open the `make_frames` file and find the line containing:

```
mo HOMO-1
```

or

```
mo HOMO
```

LUMO Generation

If the script contains:

```
mo HOMO-1
```

Jmol will generate images for the **LUMO**.

HOMO Generation

To generate images for the **HOMO**, remove the **-1** so the line becomes:

```
mo HOMO
```

5. Generate Orbital Frames

Run:

```
bash make_frames
```

Note: This process may take several minutes.

If the process stops midway through, sometimes pressing:

```
Ctrl + C
```

resolves the issue.

Once the process finishes, fix the PNG formatting:

```
bash conv_png
```

You should now see the directory populated with PNG images of your molecule(s) and orbitals.

6. Create the GIF Animation

Now that all PNG files are correctly formatted, generate the GIF.

First, initialize your Conda environment, then run:

```
python make_gif.py
```

If an error occurs indicating that a required package is missing, install it:

```
conda install NameOfMissingPackage
```

Then rerun:

```
python make_gif.py
```

You should now see a file named something like `animated_output.gif`

7. Generate the Other Orbital

Repeat **Steps 5–6** after changing the orbital selection in `make_frames`:

- `mo HOMO-1` → generates the **LUMO**
- `mo HOMO` → generates the **HOMO**

This will create the GIF for whichever orbital you have not yet generated.

8. Generate the Other Rotation Set

Repeat **Steps 2–6** for whichever rotation directory (`perp` or `para`) you did not process originally.